

Tarea Sección 5.4

5) a) $3 = \pi - 3 = 0.1416 \cdot \frac{180}{\pi} \approx \underline{8.1^\circ}$

b) $-2 = \pi - 2 = 1.1416 \cdot \frac{180}{\pi} \approx \underline{65.4^\circ}$

c) $5.5 = 2\pi - 5.5 = 0.7832 \cdot \frac{180}{\pi} \approx \underline{44.9^\circ}$

d) $100 = 32\pi - 100 = 0.5370 \cdot \frac{180}{\pi} \approx \underline{30.4^\circ}$

7) a) $\sin(2\pi/3) =$

$$\pi - \frac{2\pi}{3} = \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

b) $\sin(-5\pi/4) =$

$$\frac{5\pi}{4} - \pi = \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

77) a) $\tan(5\pi/6)$

$$\frac{5\pi}{6} - \pi = \frac{\pi}{6} = -\frac{\sqrt{3}}{3}$$

$$b) \tan(-\pi/3)$$

$$\frac{\pi}{3} = -\sqrt{3}$$

17) a) $\csc 240^\circ$

$$240 - 180 = 60^\circ \quad = -\frac{2\sqrt{3}}{3}$$

b) $\csc -330^\circ$

$$360 - 330 = 30^\circ = 2$$

23) a) $\sec 67^\circ 50'$

$$2.650$$

b) $\csc 0.32$

$$3.179$$

35) $\csc \theta = 2.54$

$$\sin^{-1}\left(\frac{1}{2.54}\right) = 23.7850 \quad \text{a)}$$

$$23^\circ 17' \quad \text{b)}$$

37) a) $\sin 98^\circ 10' = 0.9899$

b) $\cos 623.7^\circ = -0.1097$

$$c) \tan 3 = \underline{-0.1425}$$

$$f) \csc 0.82 = \underline{1.3677}$$

$$d) \cot 23^\circ 10' = \underline{0.7907}$$

$$e) \sec 1175.1 = \underline{-11.2493}$$

$$39) a) \sin \theta = -0.5640$$

$$\sin \theta = \text{III y IV}$$

$$\sin^{-1} \theta = 34.33$$

$$\theta_1 = 34.33 + 180 = 214.3^\circ$$

$$\theta_2 = 34.33 - 360 = \underline{325.7^\circ}$$

$$b) \cos \theta = 0.7490$$

$$\cos \theta = \text{I y IV}$$

$$\cos^{-1} \theta = 41.50$$

$$\theta_1 = 41.50^\circ$$

$$\theta_2 = 360 - 41.50 = \underline{318.5^\circ}$$

$$c) \tan \theta = 2.798$$

$$\tan \theta = \text{I y III}$$

$$\tan^{-1} \theta = 70.33$$

$$\theta_1 = 70.33^\circ$$

$$\theta_2 = 180 + 70.33 = \underline{250.33^\circ}$$

$$d) \cot \theta = -0.9607$$

$$\cot \theta = \text{II y IV}$$

$$\cot^{-1} \theta = 46.77$$

$$\theta_1 = 180 - 46.77 = 133.83^\circ$$

$$\theta_2 = 360 - 46.77 = \underline{313.83^\circ}$$

$$e) \sec \theta = -1.716$$

$$\sec \theta = \text{II y III}$$

$$\sec^{-1} \theta = 26.36$$

$$\theta_1 = 180 - 26.36 = 153.64$$

$$\theta_2 = 180 + 26.36 = \underline{206.4^\circ}$$

$$f) \csc \theta = 1.485$$

$$\csc \theta = \text{I y \#}$$

$$\csc^{-1} \theta = 42.33$$

$$\theta_1 = 42.33$$

$$\theta_2 = 180 - 42.33 = 137.67$$

$$41) a) \sin \theta = 0.4495$$

$$\sin \theta = \text{I y II}$$

$$\sin^{-1} \theta = 0.43$$

$$\theta_1 = 0.43$$

$$\theta_2 = \pi - 0.43 = 2.71$$

$$b) \cos \theta = -0.7207$$

$$\cos \theta = \text{II y \#}$$

$$\cos^{-1} \theta = 1.45$$

$$\theta_1 = \pi - 1.45 = 1.69$$

$$\theta_2 = \pi + 1.45 = 4.59$$

$$c) \tan \theta = -3.2504$$

$$\tan \theta = \text{II y \#}$$

$$\tan^{-1} \theta = 1.27$$

$$\theta_1 = \pi - 1.27 = 1.87$$

$$\theta_2 = 2\pi - 1.27 = 5.01$$

$$d) \cot \theta = 2.6815$$

$$\cot \theta = \text{I y III}$$

$$\cot^{-1} \theta = 0.36$$

$$\theta_1 = 0.36$$

$$\theta_2 = \pi + 0.36 = 3.50$$

$$e) \sec \theta = 1.7452$$

$$\sec \theta = \text{I y II}$$

$$\sec^{-1} \theta = 0.96$$

$$\theta_1 = 0.96$$

$$\theta_2 = 2\pi - 0.96 = 5.32$$

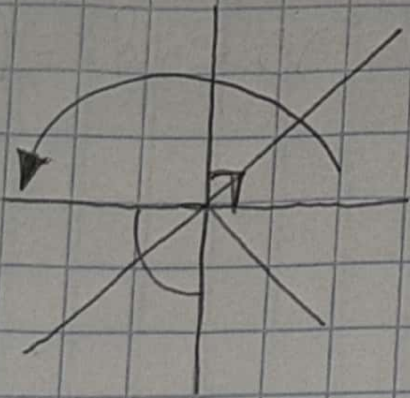
$$f) \csc \theta = -4.8521$$

$$\theta_1 = \pi + 0.27 = 3.35$$

$$\theta_2 = 2\pi - 0.27 = 6.07$$

45) Radiación solar

$$R = R_0 \cos \theta \sin \phi$$



a) El máximo ocurre cuando sale en el este.

$$b) R_0 = \cos 60^\circ \sin 45^\circ$$
$$R_0 = \frac{\sqrt{2}}{4} = 35\%$$

47) Brazo robótico

$$\sin 60^\circ \cdot 18 = 9\sqrt{3}$$

$$\cos 60^\circ \cdot 18 = 9$$

$$(9, 9\sqrt{3})$$